

Roll No.:

END SEMESTER BACK EXAMINATION 2025

Name of the Program: B.Tech

Semester: VIII

Name of the Course: Multimedia Systems and data compression

Course Code: TCS 823

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Derive Nyquist's theorem mathematically. How does quantization error impact the fidelity of digitally reconstructed audio signals?	CO1 CO3
(b)	Describe the key differences between analog and digital video signals. Explain the purpose of the NTSC, PAL, and SECAM standards in analog video transmission.	CO2
(c)	Compare MPEG-1, MPEG-2, and MPEG-4 compression techniques. How do they address temporal redundancy in video streaming?	
Q2		
(a)	Compare LZ77 (Sliding Window) and LZ78 algorithms. Illustrate how LZW improves upon these methods with an example. Why is Deflate preferred in ZIP archives?	CO4
(b)	Discuss the factors that affect the perceived quality of digital audio. Explain how bit depth, sampling rate, and quantization error influence the dynamic range and signal-to-noise ratio of audio recordings.	CO2 CO2
(c)	Compute Huffman codes for symbols [A:0.25, B:0.15, C:0.4, D:0.2]. Why is arithmetic coding more efficient for skewed distributions? Explain	
Q3		
(a)	Justify SVG over PNG for a responsive e-learning platform. Include DOM manipulation and accessibility aspects.	CO4
(b)	Diagnose color banding artifacts in a JPEG-compressed image. Propose solutions using DCT quantization tables.	CO3
(c)	Reverse-engineer the 7-zip format: How does LZMA achieve better compression than Deflate in ZIP?	CO3
Q4		
(a)	Describe the end-to-end process of creating, sending, and retrieving an MMS message containing multiple images and audio clips. How compression techniques are applied to optimize the payload size.	CO5 CO5 CO4
(b)	Explain the role of haptic interfaces in VR systems. Discuss the design considerations for creating realistic tactile feedback in surgical training simulations.	
(c)	How can MPEG-4 object-based encoding be used to optimize latency and enhance the user experience in a collaborative VR environment?	
Q5		

(a)	Explain how Discrete Cosine Transform (DCT) is used in JPEG compression. Discuss the role of quantization tables in controlling the trade-off between compression ratio and image quality.	CO5
(b)	Describe the steps involved in converting an image from RGB to YUV color space. Why is YUV preferred over RGB for video compression?	CO6
(c)	Discuss the principles behind video compression techniques such as motion estimation and motion compensation. Explain how these techniques are used in MPEG and MPEG-4 to reduce temporal redundancy in video sequences	CO6